

Regularity in the Calculus of Variations

MA 374A Assignment 1

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1. Let $s > 0$ and let

$$\Omega = B_1^n(0) := \{x \in \mathbb{R}^n : |x| < 1\} \quad \text{and} \quad u(x) = |x|^{-s}.$$

Find conditions on s such that $u \in L^q(\Omega)$ and $u \in W^{1,p}(\Omega)$. For any $1 \leq p < n$, use this to show that the Sobolev exponent p^* is sharp.

2. Let $n \geq 2$ and consider the function $u(x) = x/|x|$ in $B_1^n(0)$. Find a range for p such that $u \in W^{1,p}(B_1(0))$. This example shows Sobolev functions can be discontinuous if $n \geq 2$. Show that for $n = 1$, every Sobolev function is necessarily continuous.
3. Show that for any $0 < s < 1/2$, $u(x) = |\log |x||^s \in W^{1,2}(B_{1/2}^2(0))$.
4. Show that $u(x) = \log \log(1 + 1/|x|) \in W^{1,n}(B_1^n(0))$ for $n \geq 2$.
5. Use the Sobolev representation/local Poincaré inequality to prove the following.

Theorem 1. *For every $1 \leq p < \infty$, there exists a constant $c = c(n, p) > 0$ such that*

$$\oint_{B_R(x)} \left| u - (u)_{B_R(x)} \right|^p \leq c r^p \oint_{B_R(x)} |\nabla u|^p$$

for any ball $B_R(x) \subset \mathbb{R}^n$ and every $u \in W^{1,p}(\mathbb{R}^n)$.

6. Provide a constructive proof for the following.

Theorem 2. Let $\Omega \subset \mathbb{R}^n$ be open, bounded and C^1 and convex. For every $1 \leq p < \infty$, there exists a constant $c = c(n, p, \Omega) > 0$ such that

$$\int_{\Omega} |u - (u)_{\Omega}|^p \leq c \int_{\Omega} |\nabla u|^p$$

for every $u \in W^{1,p}(\Omega)$.

Remark 3. The result holds without the convexity hypothesis, as shown below. Think about how you can find a constructive proof.

7. Use Rellich-Kondrachov compactness to establish the following.

Theorem 4. Let $\Omega \subset \mathbb{R}^n$ be open, bounded and C^1 . For every $1 \leq p < \infty$, there exists a constant $c = c(n, p, \Omega) > 0$ such that

$$\int_{\Omega} |u - (u)_{\Omega}|^p \leq c \int_{\Omega} |\nabla u|^p$$

for every $u \in W^{1,p}(\Omega)$.

8. Let $\Omega \subset \mathbb{R}^n$ be open, bounded and C^1 . Show that

$$W^{1,\infty}(\Omega) = C^{0,1}(\overline{\Omega}) \quad (\text{with equivalent norms}).$$

9. Prove that for any measurable set $A \subset \mathbb{R}^n$ with $0 < |A| < \infty$ and any $1 \leq p < \infty$, we have

$$\int_A |f - (f)_A|^p \leq 2 \int_A |f - \lambda|^p \quad \text{for every } \lambda \in \mathbb{R} \text{ for any } f \in L^p(A).$$

Remark 5. For $p = 2$, the constant can be improved to 1.